Language Models (LMs)

Natural Language Processing

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Language modeling

- Language modeling refers to different probabilistic and statistical methods used to determine the chances of a given series of words occurring in a text. These models in language modeling analyze text and provide predictions as to which words might come next.
- https://www.xavor.com/blog/language-modeling-in-nlp-a-comprehensive-overview/

How it works

1. Training:
   - Language models are trained on vast datasets like Wikipedia, books, or news articles. The model learns to understand the structure and patterns of the language from this data.
2. Prediction Process:
   - When a model is given a sequence of words, it looks at the past words (context) and predicts the next word by selecting the one with the highest probability.
3. Evaluation:
   - The performance of language models is often measured using a metric called **perplexity**, which indicates how well the model predicts the next word. Lower perplexity means better performance.

Language Models

- Models that assign probabilities to sequences of words
- Assigns a probability to each possible next word (or sentence).
- Predicts the next word given a context
  - "he briefed reporters on the main contents of the <next word?>"
- Which of the following is more likely?
  1. "all of a sudden I notice three guys standing on the sidewalk"
  2. "on guys all I of notice sidewalk three a sudden standing the"

Corpus statistics

i	want	to	eat	chinese	food	lunch	spend
2533	927	2417	746	158	1093	341	278

Example

	i	want	to	eat	chinese	food	lunch	spend
i	5	827	0	9	0	0	0	2
want	2	0	608	1	6	6	5	1
to	2	0	4	686	2	0	6	211
eat	0	0	2	0	16	2	42	0
chinese	1	0	0	0	0	82	1	0
food	15	0	15	0	1	4	0	0
lunch	2	0	0	0	0	1	0	0
spend	1	0	1	0	0	0	0	0

Figure 3.1 Bigram counts for eight of the words (out of $V = 1446$) in the Berkeley Restaurant Project corpus of 9332 sentences. Zero counts are in gray.

Example (probabilities)

	i	want	to	eat	chinese	food	lunch	spend
i	0.002	0.33	0	0.0036	0	0	0	0.00079
want	0.0022	0	0.66	0.0011	0.0065	0.0065	0.0054	0.0011
to	0.00083	0	0.0017	0.28	0.00083	0	0.0025	0.087
eat	0	0	0.0027	0	0.021	0.0027	0.056	0
chinese	0.0063	0	0	0	0	0.52	0.0063	0
food	0.014	0	0.014	0	0.00092	0.0037	0	0
lunch	0.0059	0	0	0	0	0.0029	0	0
spend	0.0036	0	0.0036	0	0	0	0	0

Figure 3.2 Bigram probabilities for eight words in the Berkeley Restaurant Project corpus of 9332 sentences. Zero probabilities are in gray.

Types

- N-gram
- Neural language models

Applications

- Machine Translation
- Spell Correction
- Speech Recognition
- Text Generation

Some terminologies

- Unigram
  - A word/token
- Bigram
  - two-word sequence
- Trigram
  - three word sequence
- N-gram
  - Sequence of N words

Corpus statistics

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Example: Bigram counts

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Bigram probabilities

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N-gram model

Simplistic approach

- computing P(w|h), the probability of a word w given some history h.
  - P(the | its water is so transparent that)
  - $\frac{C(its\ water\ is\ so\ transparent\ that\ the)}{C(its\ water\ is\ so\ transparent\ that)}$
- even the web isn’t big enough to give us good estimates in most cases.
- new sentences are created all the time, and we won’t always be able to count entire sentences.
  - any particular context might have never occurred before!

Chain rule of probability

$$
\begin{aligned}
P(X_1 \dots X_n) &= P(X_1)P(X_2|X_1)P(X_3|X_1^2)\dots P(X_n|X_1^{n-1}) \\
&= \prod_{k=1}^{n} P(X_k|X_1^{k-1}) \qquad (3.3)
\end{aligned}
$$

Applying the chain rule to words, we get

$$\begin{aligned} P(w_1^n) &= P(w_1)P(w_2|w_1)P(w_3|w_1^2)\ldots P(w_n|w_1^{n-1}) \\ &= \prod_{k=1}^{n} P(w_k|w_1^{k-1}) \end{aligned} \quad (3.4)$$

- We don’t know any way to compute the exact probability of a word given a long sequence of preceding words, P($w_n | w_1^{n-1}$).
- we can’t just estimate by counting the number of times every word occurs following every long string, because language is creative and any particular context might have never occurred before!

Solution (n-gram)

- instead of computing the probability of a word given its entire history, we can approximate the history by just the last few words.

Bigram model

- The bigram model, for example, approximates the probability of a word given all the previous words P(w_n | w_1^{n-1}) by using only the conditional probability of the preceding word P(w_n | w_{n-1}).
- Instead of computing P(the | its water is so transparent that)
- We compute P(the | that)

$$P(w_n|w_1^{n-1}) \approx P(w_n|w_{n-1})$$

Markov model

- The **assumption** that the probability of a word depends only on the previous word Markov is called a **Markov assumption**.
- Markov models are the class of probabilistic models that assume we can predict the probability of some future unit without looking too far into the past. We can generalize the bigram (which looks one word into the past) to the trigram (which looks two words into the past) and thus to the n-gram (which looks $n-1$ words into the past).

Generalized formula

$$P(w_n|w_1^{n-1}) \approx P(w_n|w_{n-N+1}^{n-1})$$

What is N here?

Generalized formula (Bigram)

$$P(w_1^n) \approx \prod_{k=1}^{n} P(w_k | w_{k-1})$$

Exercise

- P(I|<S>)?
- P(Sam|am)?

Corpus:

```
<s> I am Sam </s>
<s> Sam I am </s>
<s> I do not like green eggs and ham </s>
```

$$P(w_n|w_{n-1}) = \frac{C(w_{n-1}w_n)}{C(w_{n-1})}$$

```
<s> I am Sam </s>
<s> Sam I am </s>
<s> I do not like green eggs and ham </s>
```

Here are the calculations for some of the bigram probabilities from this corpus

$$P(\texttt{I}\,|\,\texttt{<s>}) = \tfrac{2}{3} = .67 \qquad P(\texttt{Sam}\,|\,\texttt{<s>}) = \tfrac{1}{3} = .33 \qquad P(\texttt{am}\,|\,\texttt{I}) = \tfrac{2}{3} = .67$$

$$P(\texttt{</s>}\,|\,\texttt{Sam}) = \tfrac{1}{2} = 0.5 \qquad P(\texttt{Sam}\,|\,\texttt{am}) = \tfrac{1}{2} = .5 \qquad P(\texttt{do}\,|\,\texttt{I}) = \tfrac{1}{3} = .33$$

For the general case of MLE n-gram parameter estimation:

$$P(w_n|w_{n-N+1}^{n-1}) = \frac{C(w_{n-N+1}^{n-1}w_n)}{C(w_{n-N+1}^{n-1})} \qquad (3$$

Generation capabilities comparison

1 gram	–To him swallowed confess hear both. Which. Of save on trail for are ay device and rote life have –Hill he late speaks; or! a more to leg less first you enter
2 gram	–Why dost stand forth thy canopy, forsooth; he is this palpable hit the King Henry. Live king. Follow. –What means, sir. I confess she? then all sorts, he is trim, captain.
3 gram	–Fly, and will rid me these news of price. Therefore the sadness of parting, as they say, ’tis done. –This shall forbid it should be branded, if renown made it empty.
4 gram	–King Henry. What! I will go seek the traitor Gloucester. Exeunt some of the watch. A great banquet serv’d in; –It cannot be but so.

Figure 3.3 Eight sentences randomly generated from four n-grams computed from Shakespeare’s works. All characters were mapped to lower-case and punctuation marks were treated as words. Output is hand-corrected for capitalization to improve readability.

1 gram	Months the my and issue of year foreign new exchange's september were recession exchange new endorsed a acquire to six executives
2 gram	Last December through the way to preserve the Hudson corporation N. B. E. C. Taylor would seem to complete the major central planners one point five percent of U. S. E. has already old M. X. corporation of living on information such as more frequently fishing to keep her
3 gram	They also point to ninety nine point six billion dollars from two hundred four oh six three percent of the rates of interest stores as Mexico and Brazil on market conditions

Figure 3.4 Three sentences randomly generated from three n-gram models computed from 40 million words of the *Wall Street Journal*, lower-casing all characters and treating punctuation as words. Output was then hand-corrected for capitalization to improve readability.

Quiz (Reading)

- What is Hidden Markov Model. Explain its importance with the help of an example?
- What are transition probabilities?